

Meeting01

Minutes of the 1st Project Meeting: one hour

- Date: 20 Sep. 2025 (Saturday)
- Time: 16:00 pm
- Place: Online Meeting
- Attending:
 - Mentor: HU, Siyan
 - Student: CHANG, Ruihe
- Absent: None
- Recorder: CHANG, Ruihe

Approval of minutes

- The minutes of the meeting were reviewed and approved by HU, Siyan.

Discussion items

- Refactor the entire MSBD5014A project using React and TypeScript.
 - Plan to implement new technologies and optimization strategies to achieve faster response times, enhanced rendering quality, and lower resource consumption.
- Outline the new features to be implemented.
 - Utilize Large Language Models and Server-Sent Events to enable real-time AI communication and point file analysis.

Meeting adjournment and next meeting

- The meeting was adjourned at 17:00 PM. The next meeting will be held in October.

Meeting02

Minutes of the 2nd Project Meeting: one hour

- Date: 28 Oct. 2025 (Tuesday)
- Time: 16:40 pm
- Place: Online Meeting
- Attending:
 - Mentor: HU, Siyan
 - Student: CHANG, Ruihe
- Absent: None
- Recorder: CHANG, Ruihe

Approval of minutes

- The minutes of the meeting were reviewed and approved by HU, Siyan.

Discussion items

- Complete a full project refactor with redesigned UI, new technologies, and advanced optimization strategies.
- Visualization Performance Optimizations:
 - Refresh Rate Synchronization
 - Background Resource Saving
 - Geometry Batching
 - Performance Optimizations
 - Interactive Optimizations
- Point Cloud Down-Sampling Strategies:
 - Spatial Hash Algorithm
 - Ratio-Based Down-Sampling

Meeting adjournment and next meeting

- The meeting was adjourned at 17:40 PM. The next meeting will be held in November.

Meeting03

Minutes of the 3st Project Meeting: one hour

- Date: 28 Nov. 2025 (Friday)
- Time: 20:30 pm
- Place: Online Meeting
- Attending:
 - Mentor: HU, Siyan
 - Student: CHANG, Ruihe
- Absent: None
- Recorder: CHANG, Ruihe

Approval of minutes

- The minutes of the meeting were reviewed and approved by HU, Siyan.

Discussion items

- Complete Point Cloud File Analysis Feature
 - Utilized Large Language Models to perform in-depth analysis on uploaded point cloud files.
- Real-Time AI Communication:
 - Establish Server-Sent Events (SSE) connections to enable robust data streaming.
 - Achieve real-time, token-by-token streaming output.
- Interaction Optimization:
 - Implement Debounce: Apply debounce mechanisms to prevent browser overload caused by excessive re-rendering.
 - Markdown Pre-processing: Pre-normalized Markdown symbols to eliminate UI flickering during

format parsing.

Meeting adjournment and next meeting

- The meeting was adjourned at 21:30 PM. The next meeting will be held in December.

Meeting 04

Minutes of the 4st Project Meeting: one hour

- Date: 13 Dec. 2025 (Saturday)
- Time: 17:40 pm
- Place: Online Meeting
- Attending:
 - Mentor: HU, Siyan
 - Student: CHANG, Ruihe
- Absent: None
- Recorder: CHANG, Ruihe

Approval of minutes

- The minutes of the meeting were reviewed and approved by HU, Siyan.

Discussion items

- Present the complete project and Overview all features.
- Revise the final report, reducing the description of libraries, and emphasizing the theories.
- (For more details, please see `Final Report`)

Meeting adjournment and next meeting

- The meeting was adjourned at 18:40 PM. This is the final meeting.